🌍 Climate Change AI: CO_2 Emission Prediction

1️⃣ Raw Dataset Overview

Rows: `17272`

Columns: `50`

	country	date	agricultural_land%	forest_land%	land_area	avg_precipitation	trade_in_services%	control_of_corruption_estimate	control_of_
0	Afghanistan	1960-01-01	None	None	None	None	None	None	
1	Afghanistan	1961-01-01	57.8784	None	652230	327	None	None	
2	Afghanistan	1962-01-01	57.955	None	652230	327	None	None	
3	Afghanistan	1963-01-01	58.0317	None	652230	327	None	None	
4	Afghanistan	1964-01-01	58.116	None	652230	327	None	None	

2️⃣ Cleaned Dataset

Countries: `135`

Years: `30`

Rows: `3207`

	Country	Year	Precipitation	Forest_Land	Agricultural_Land	Electric_Power	CO2	Population_Density	Population	GDP
222	Albania	1990-01-01	0.4553	0.2927	0.4758	0.0098	0.0006	0.0154	0.0022	0.00009
223	Albania	1991-01-01	0.4553	0.292	0.4784	0.0074	0.0004	0.0153	0.0022	0.00004
224	Albania	1992-01-01	0.4553	0.2913	0.4784	0.008	0.0002	0.0152	0.0022	0.00001
225	Albania	1993-01-01	0.4553	0.2906	0.478	0.0095	0.0002	0.0151	0.0022	0.00004
226	Albania	1994-01-01	0.4553	0.2899	0.478	0.0106	0.0002	0.015	0.0022	0.00008

3️⃣ Model Training

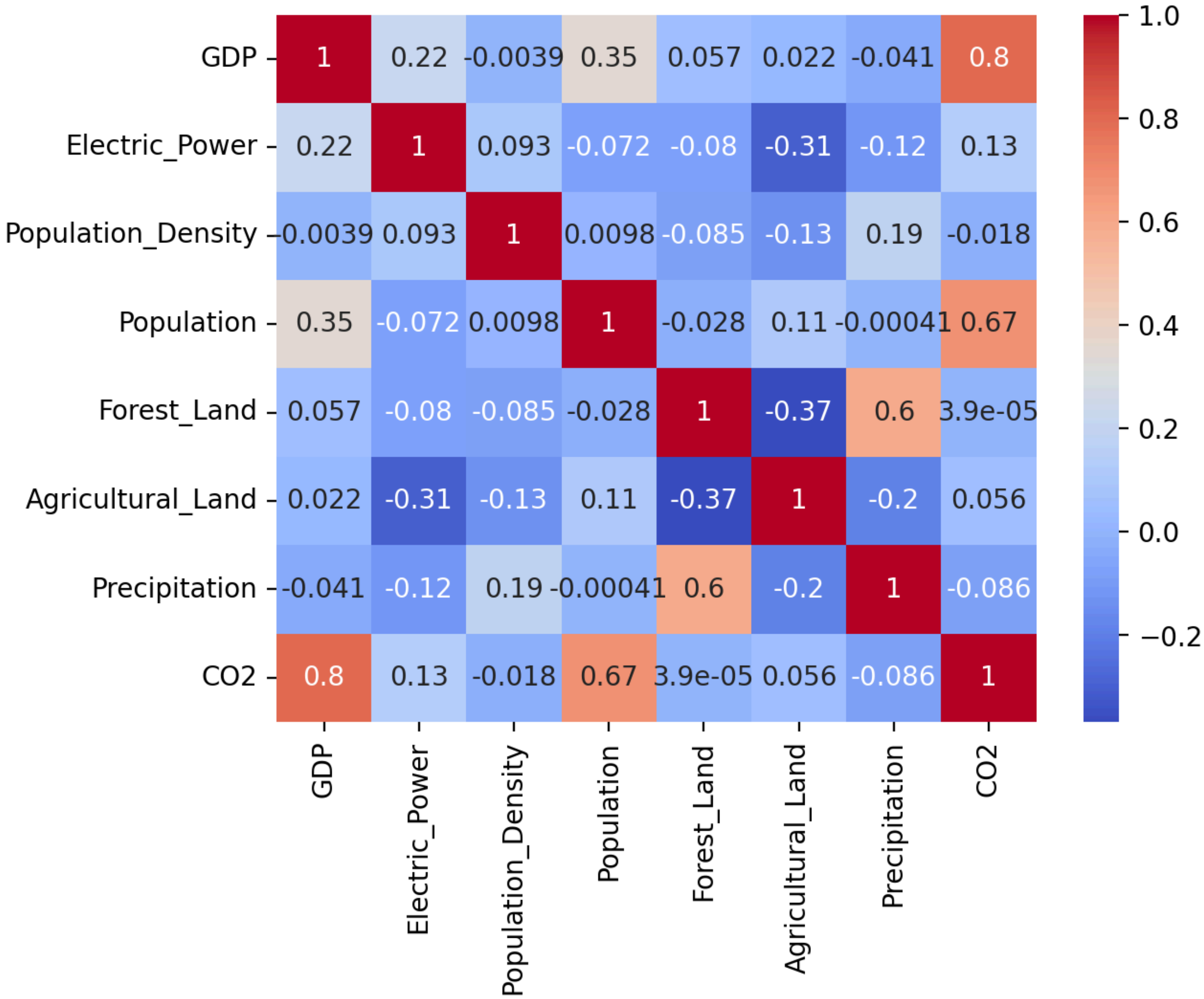
Models trained successfully!

4️⃣ Model Performance

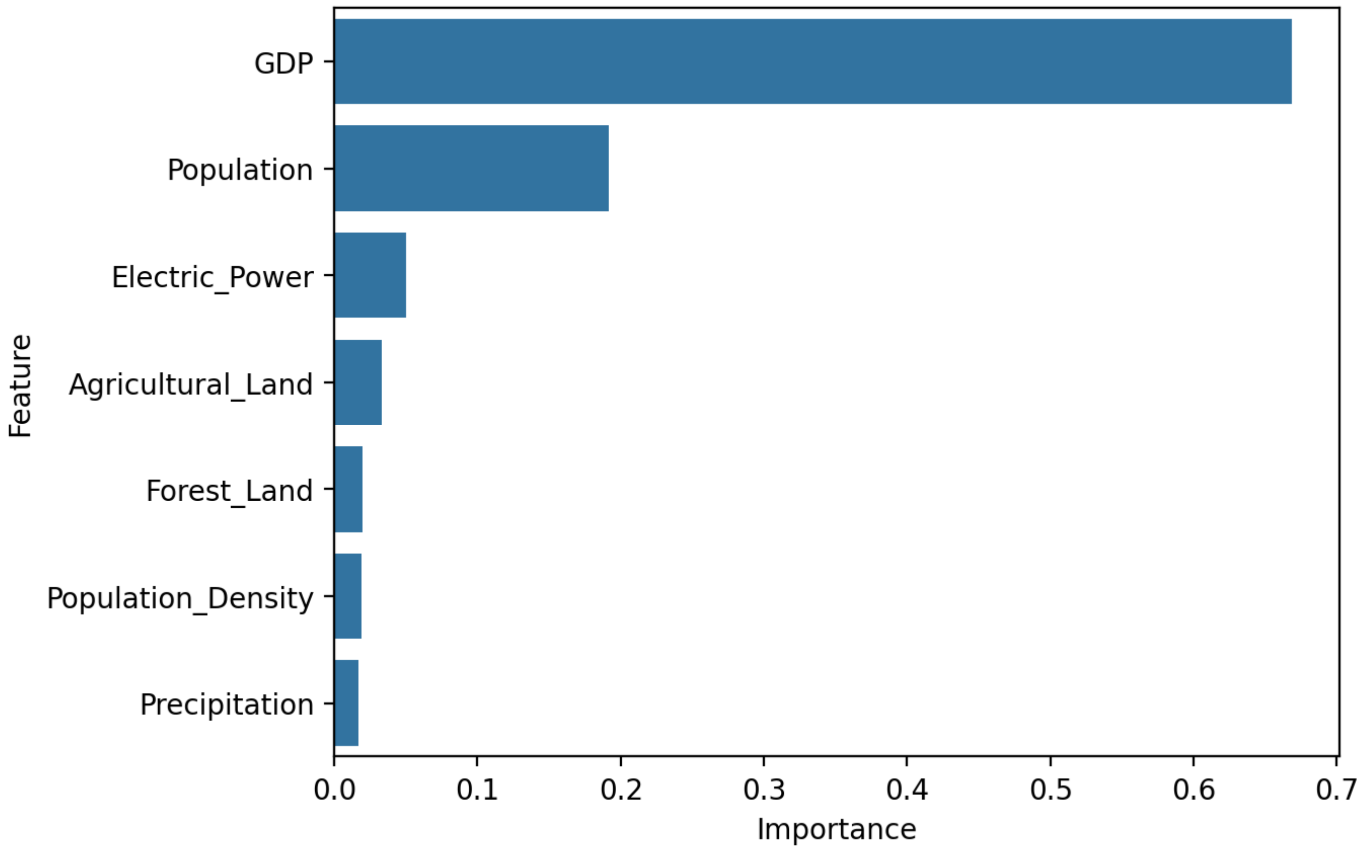
	Model	RMSE	MAE	R2 Score
0	Linear Regression	0.0354	0.0125	0.8318
1	Random Forest	0.0106	0.0019	0.9848

5️⃣ Visualizations

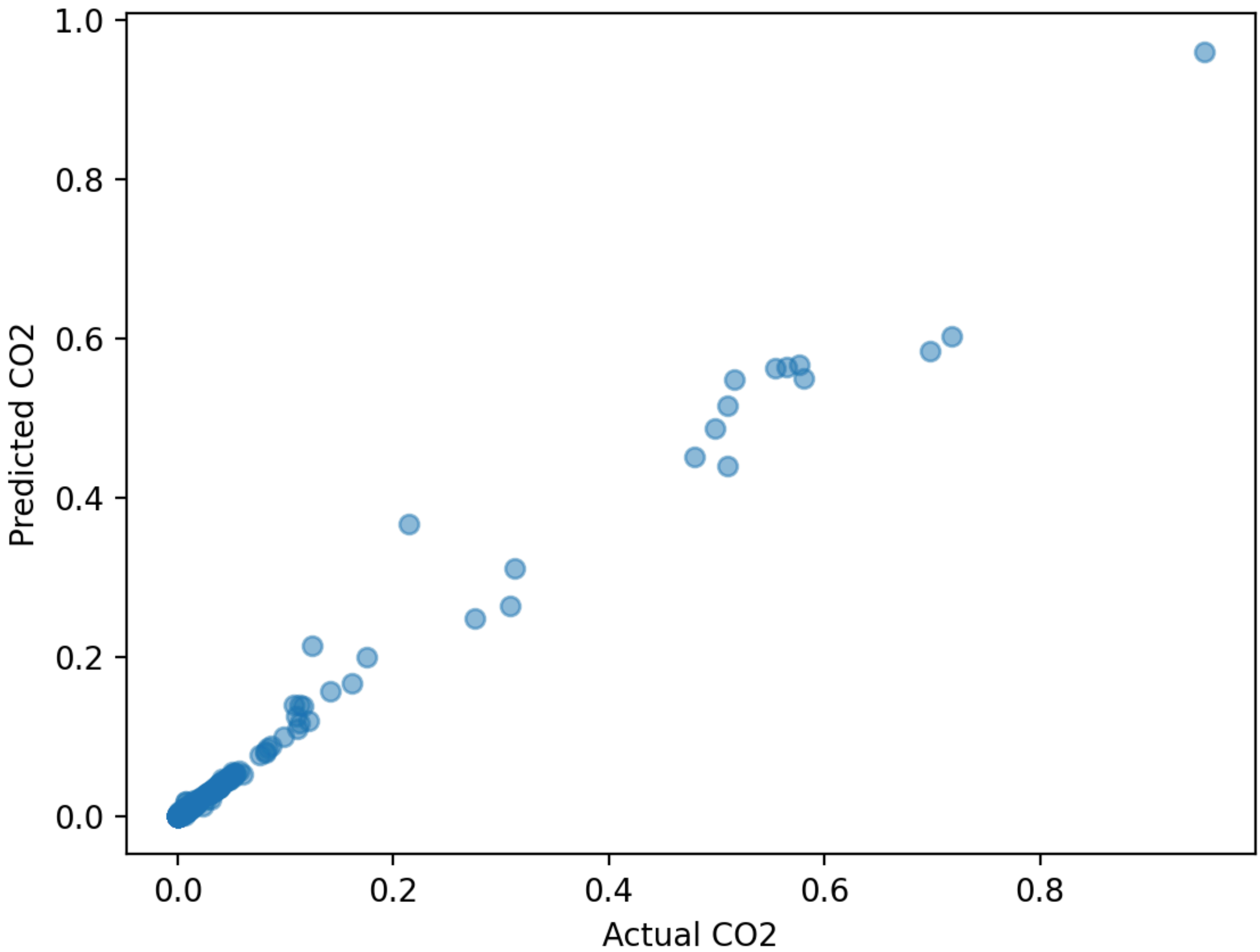
Correlation Heatmap



Feature Importance



Actual vs Predicted CO_2



CO_2 Emission Trend (Country-wise)

Select Country

India